PRESENTED BY
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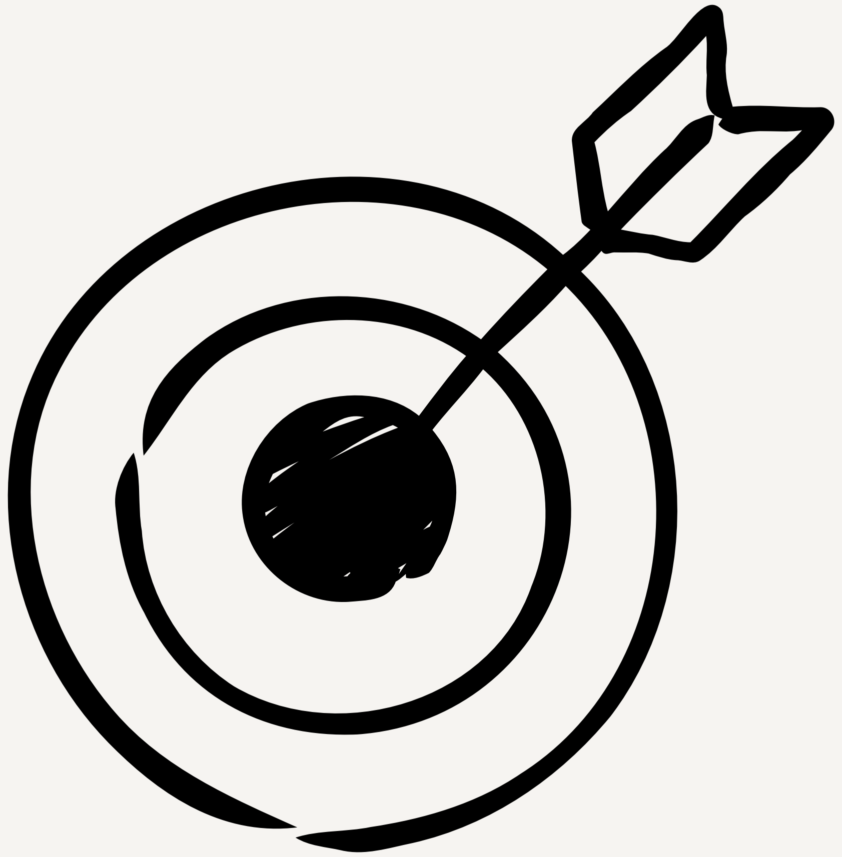
DATA-DRIVEN PROPERTY VALUATION FACTORS AND MODEL FOR CEBU CITY

A Comparative Analysis of Hedonic Regression and Machine Learning Approaches

INTRODUCTION



REAL ESTATE IMPORTANCE

Real estate is a **key driver** of the Philippine economy (~6% of GDP; PSA, 2024).

THE PROBLEM:

Property valuation in Cebu is currently **fragmented**:

- **BIR Zonal Values**: Outdated and often lower than market value.
- **Bank Appraisals**: Conservative and opaque.
- **Market Listings**: Highly variable asking prices.

THE NEED:

A transparent, data-driven model to estimate "Fair Market Value" for stakeholders like Cebu Premiere Real Estate (CPRE).

PROBLEM STATEMENT

There is no single, scientifically validated model to estimate residential property values in Cebu City.

Stakeholders rely on intuition or disjointed data sources (Agosto, 2017)

Specifically:

"How can we accurately predict the market value of a residential property in Cebu using available data?"

RESEARCH QUESTIONS

Model Comparison

- Does a Machine Learning model (Random Forest / XGBoost) **outperform traditional Hedonic Regression** in predicting property prices?

Key Drivers

- What are the **most significant determinants** of property value in Cebu (e.g., Floor Area vs. Location vs. Proximity to CBD)?

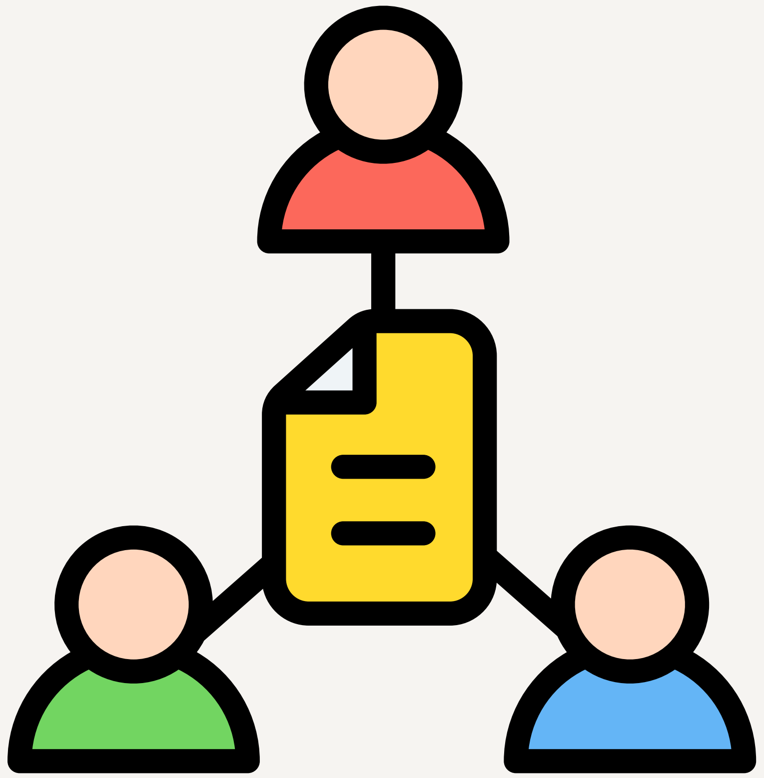
Valuation Gap

- What is the magnitude of the difference between **BIR Zonal Values** and **actual Market Asking Prices**?

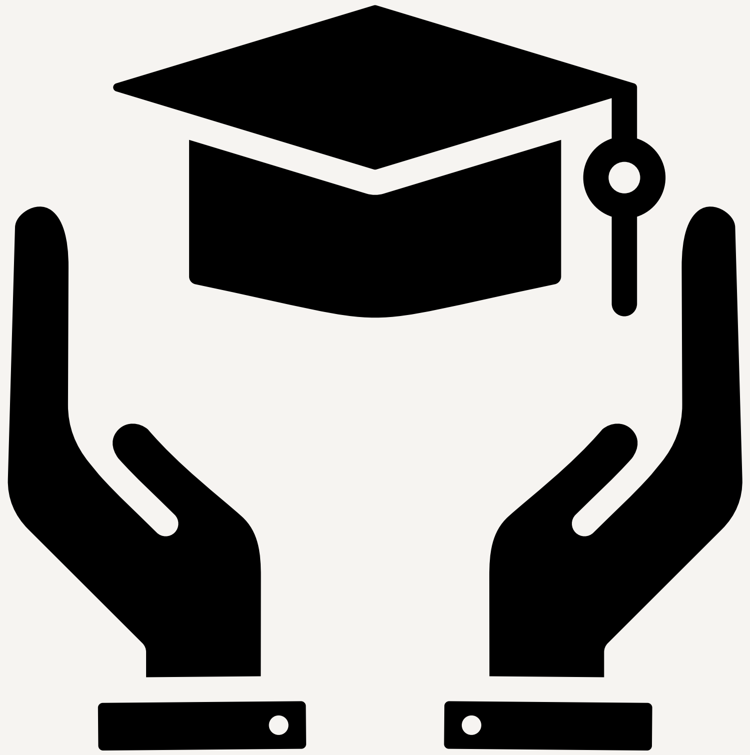
Future Factors

- How do **planned infrastructure projects** (Cebu Bus Rapid Transit) **capture value** in the model?

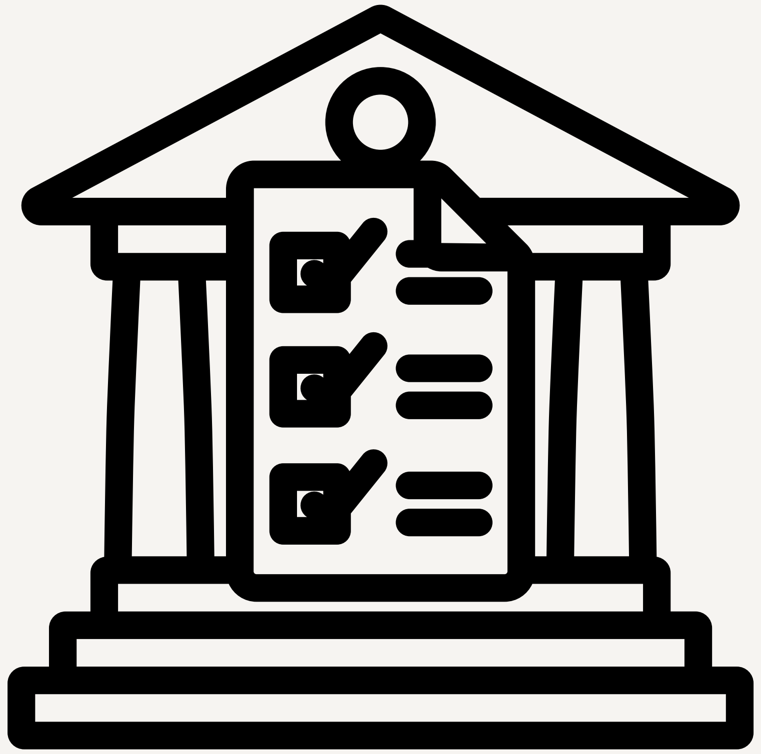
RESEARCH SIGNIFICANCE



STAKEHOLDER



ACADEME

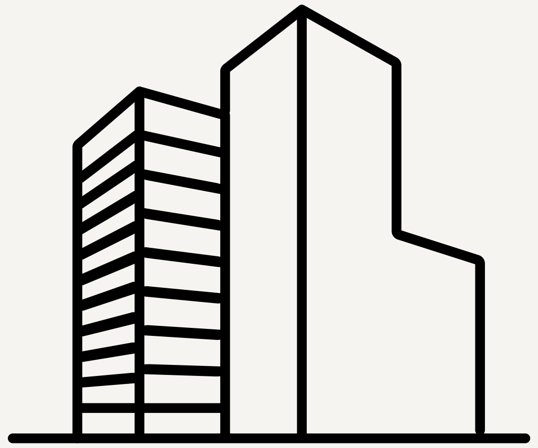


POLICYMAKERS

SCOPE AND LIMITATIONS

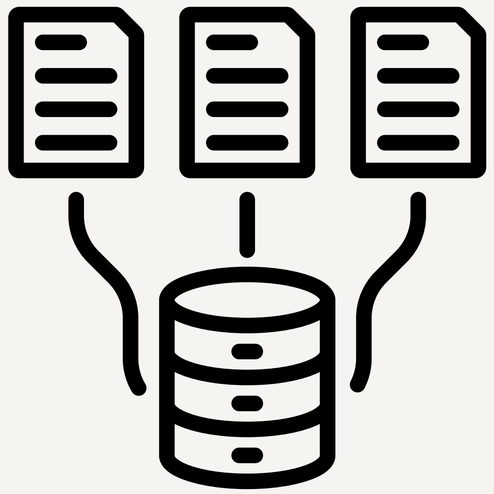
Geography

Primarily Cebu City and immediate Metro Cebu neighbors



Property Type

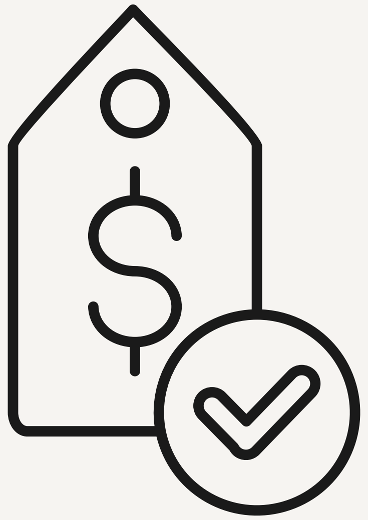
Residential (House & Lot, Condominium, Vacant Residential Lot)



Data Source

Foreclosed assets and Online Listings

REVIEW OF RELATED LITERATURE
(Theories)



Hedonic Pricing Theory (Rosen, 1974)

Value is the sum of its parts (Attributes)

P = f(Physical,Locational,Environmental)

Spatial Economics

"Accessibility" is the primary determinant of land rent (Agosto, 2017)

REVIEW OF RELATED LITERATURE
(Empirical)

Viray (2023 - Philippines)

Random Forest reduced error by 15% compared to Linear Regression in provincial valuations.

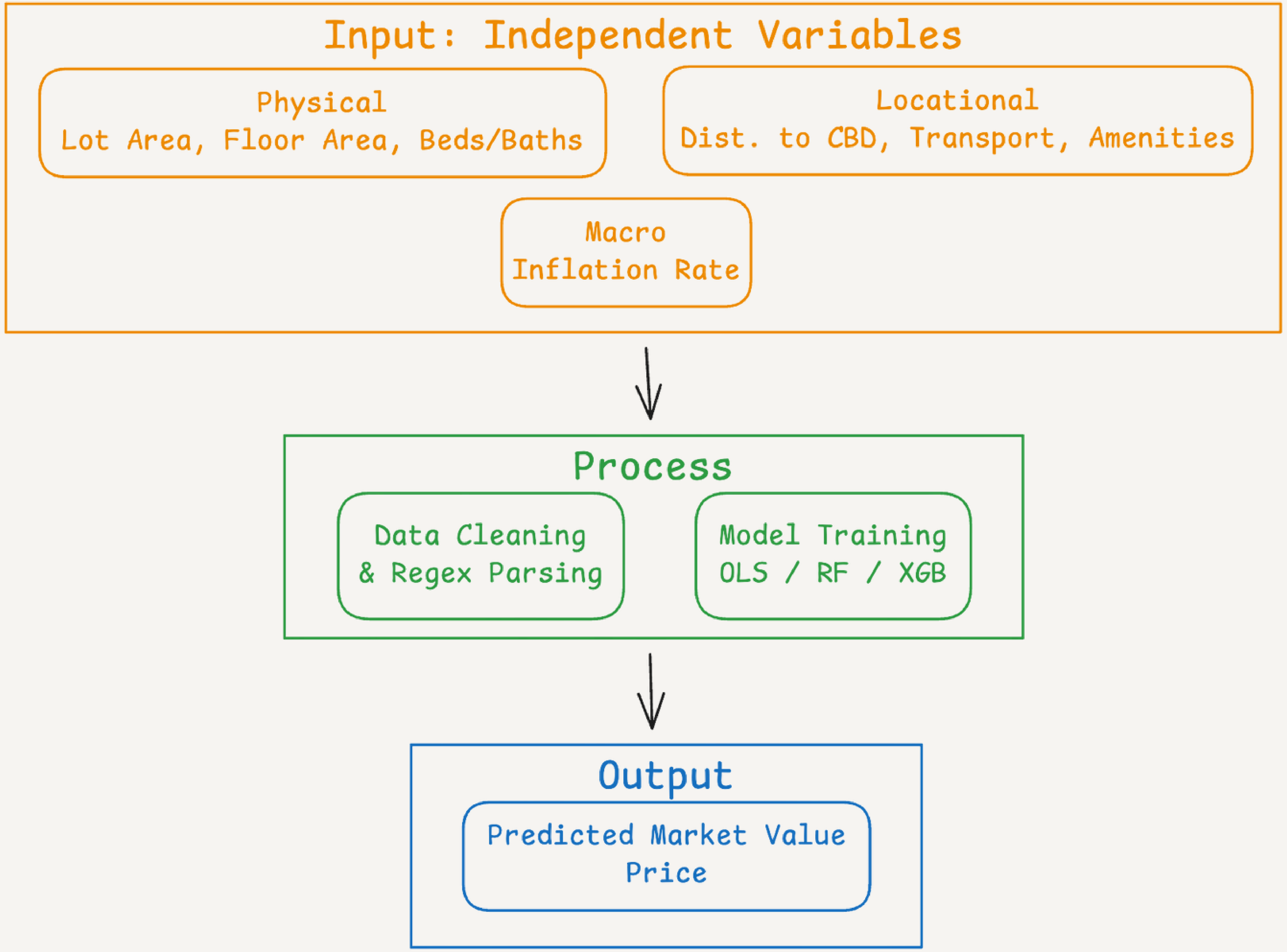
Domingo & Fulleros (2005 - Policy)

Established the "**Valuation Gap**" concept in the Philippine setting.

Gayathri & Thekkayil (2025 - International)

Ensemble Models (RF + XGBoost) achieved 97% accuracy; "Future Infrastructure" is a critical predictor

CONCEPTUAL FRAMEWORK



METHODOLOGY - DATA PROCESSING

Data Collection

- **Hybrid Data Collection:**
  - **Floor Price (Verified):** Administrative data may be undervalued (Domingo & Fulleros, 2005).
  - **Ceiling Price (Market):** Online listings effectively capture market segments, as validated by Sousa et al. (2024), even without official transaction data.
  - **The Validation Layer (Expert):** Integrate a **Human-in-the-Loop** approach

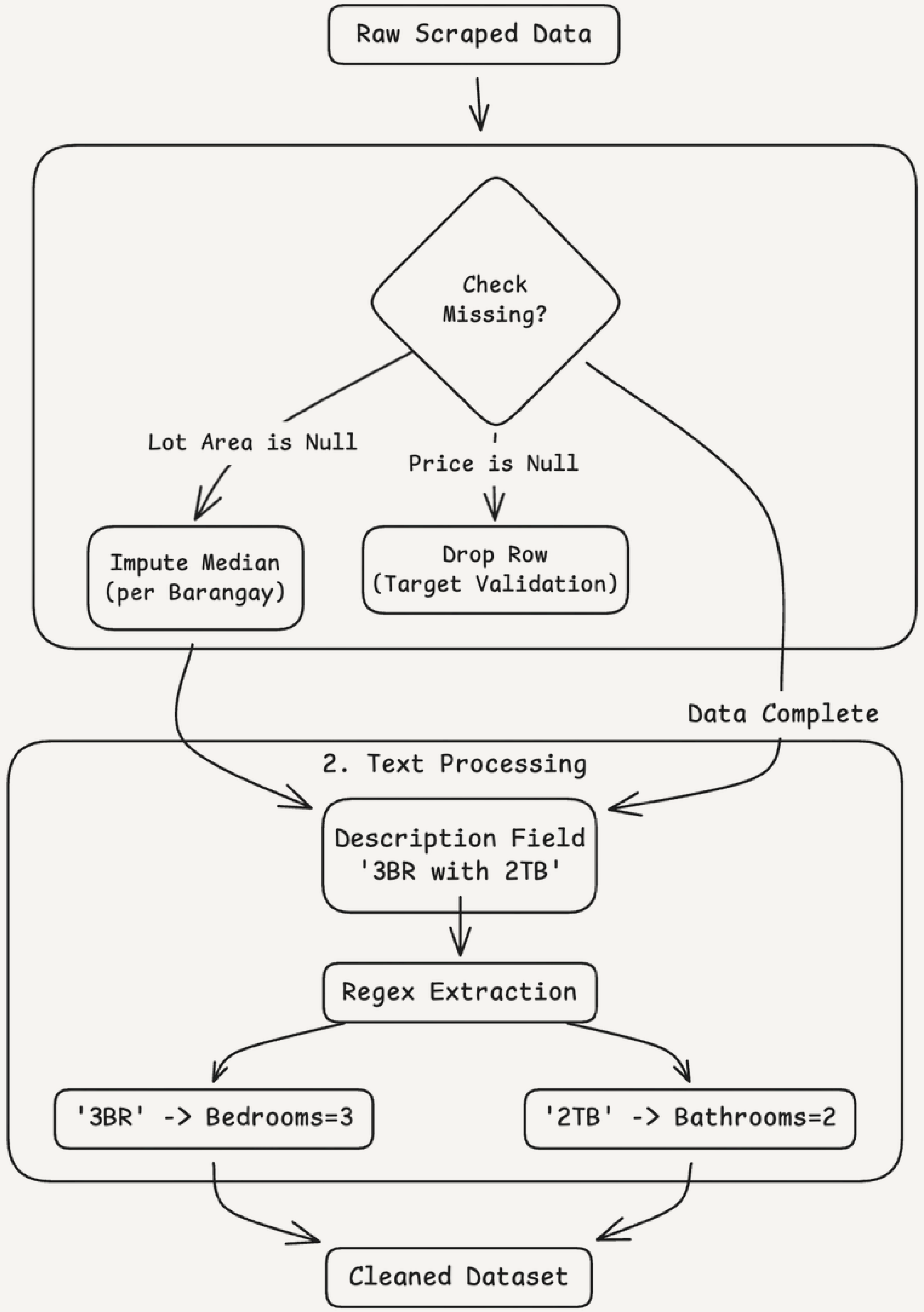
Preprocessing

- **Text Mining**: Extracting "3 Bedroom" from description fields.
- **Geocoding**: Converting addresses to Latitude/Longitude using Google Maps/OpenStreet API.

Preprocessing

- 80% Training, 20% Testing (**Stratified by City**)

DATA CLEANING STRATEGY



FEATURE ENGINEERING
(Geospatial)

We go beyond "City" labels by calculating precise vectors:

Distance to CBD (Central Business District)

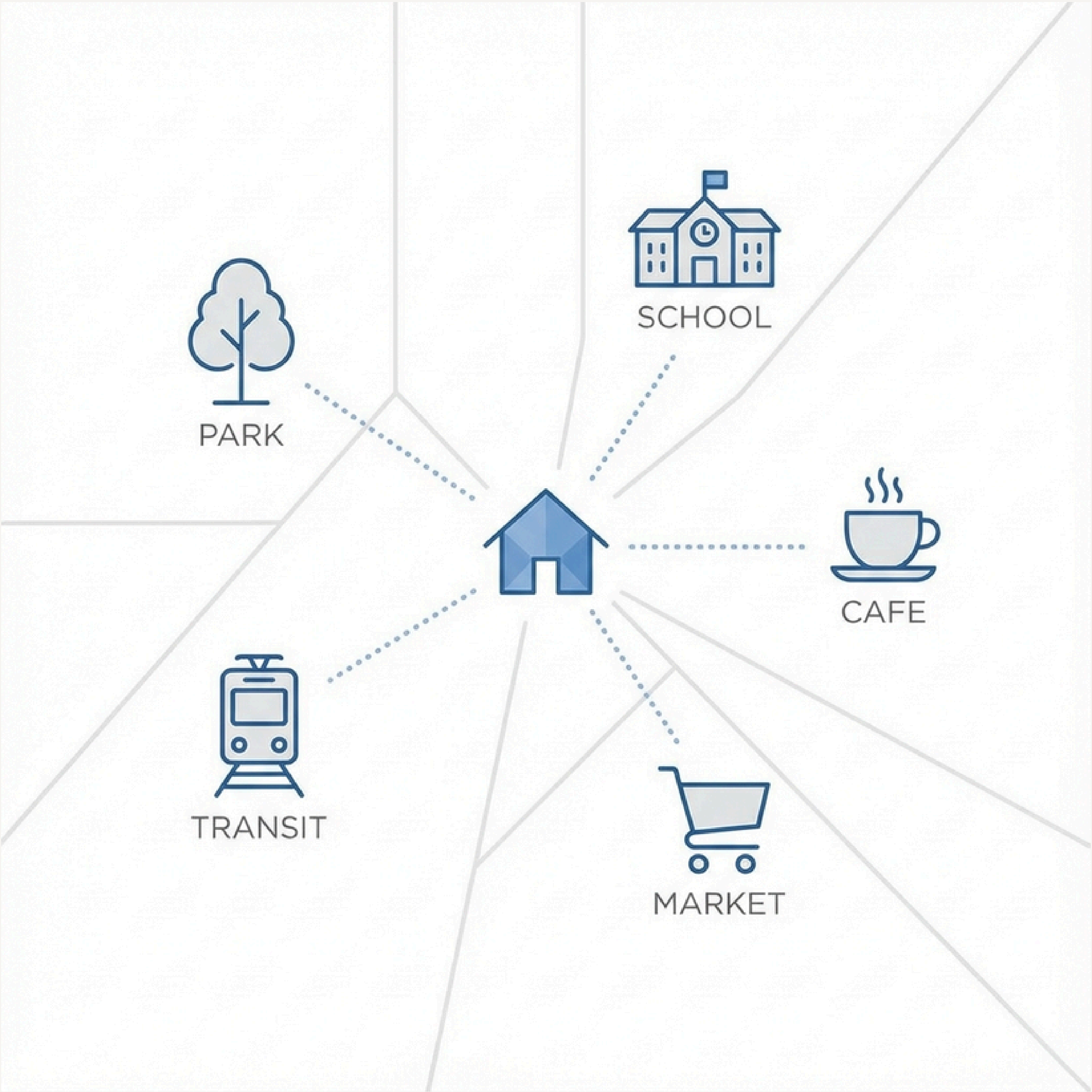
Haversine distance to Ayala Center Cebu & IT Park.

Distance to Amenities

Radius count of Schools and Hospitals within 1km.

Distance to Transport

Proximity to nearest future **CBRT Station**



METHODOLOGY - MODELING

Baseline: Ordinary Least Squares (OLS) Regression (Rosen, 1974)

- **Pros**: Highly interpretable coefficients.
- **Cons**: Assumes linearity; fails with complex interactions.

Challenger 1: Random Forest Regressor (Breiman, 2001; Viray, 2023)

Pros: Handles non-linearities and outliers well
Cons: Computationally expensive; "Black box" nature makes interpretation harder than OLS.

Challenger 2: XGBoost (Extreme Gradient Boosting)

(Chen & Guestrin, 2016; Gayathri & Thekkayil, 2025)

Pros: State-of-the-art accuracy; Gradient-based optimization
Cons: Complex hyper-parameter tuning required; sensitive to noise/outliers if not regularized.

EMPIRICAL FRAMEWORK

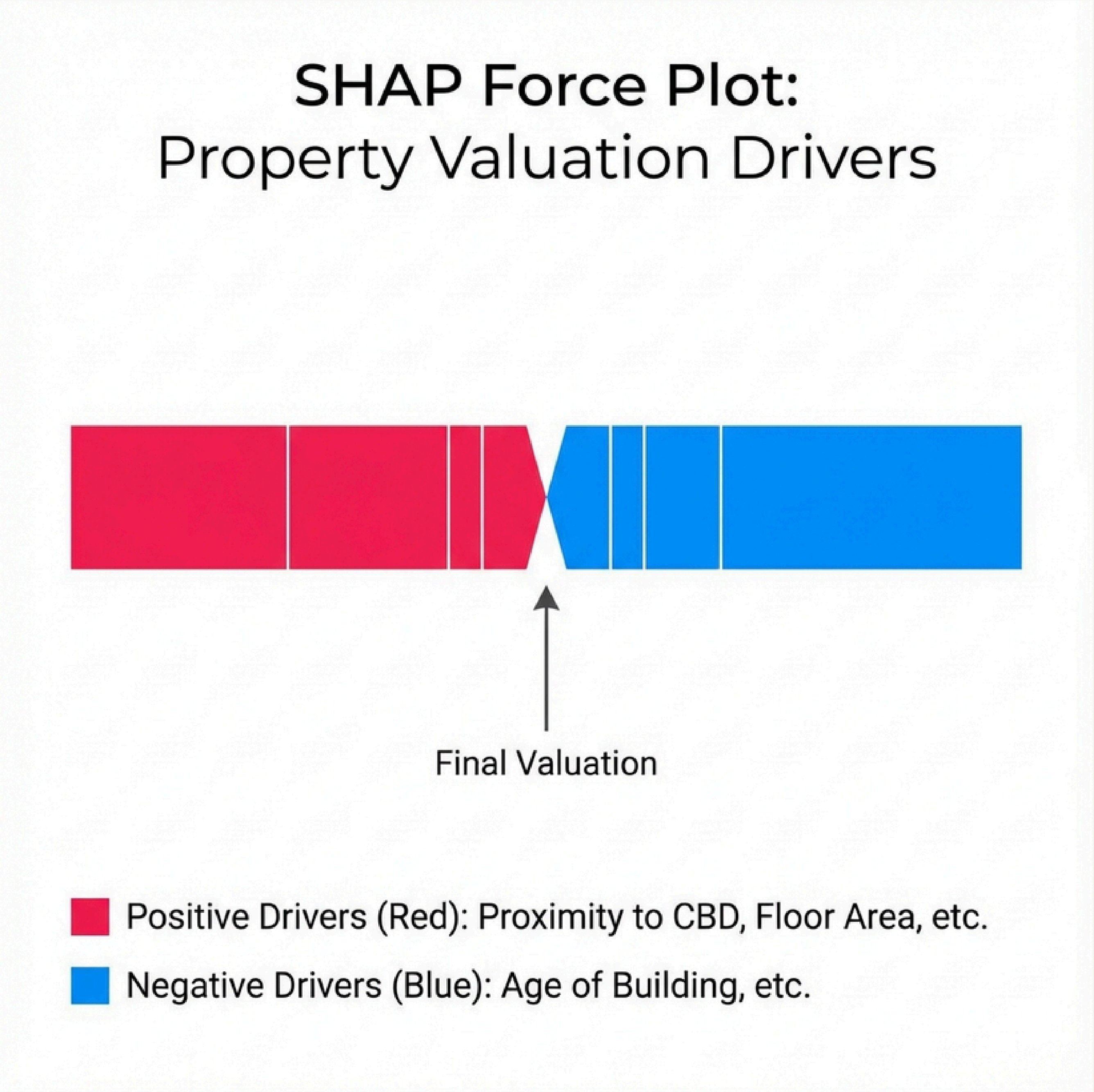
(Evaluation)

To evaluate success, we use:

- **MAE** (Mean Absolute Error):
  - most practical metric for stakeholders. It gives us the average error in **Pesos**.
  - For example, 'The model is off by +/- 500,000 pesos on average.
- **RMSE** (Root Mean Squared Error):
  - Penalizes large errors more heavily. ensures our model isn't making catastrophic mistakes on expensive properties
- **R-Squared** (r-squared):
  - How much variance in price is explained by our features? (Target: > **0.80**).

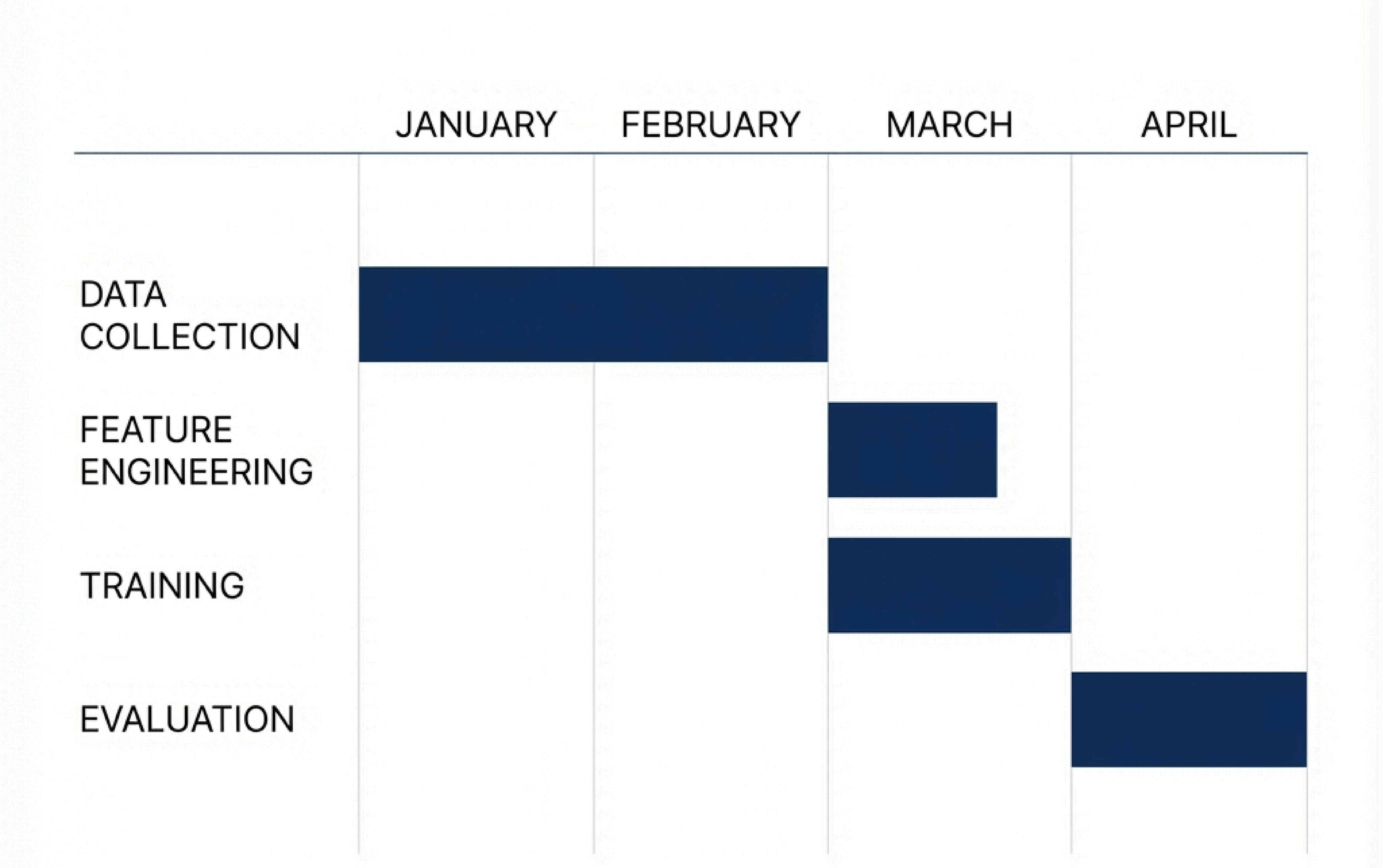
MODEL INTERPRETATION STRATEGY

- **Global Interpretability**:
  - **Feature Importance plots** (Random Forest) to identify market drivers (e.g., "Distance to CBD" vs. "Lot Area")
- **Local Interpretability:**
  - **SHAP** (Shapley Additive Explanations) values to explain individual predictions
  - **Example**: "This specific unit is priced +P500k due to 'Proximity to IT Park'."



- **Builds trust with stakeholders (CPRE) who may be skeptical of "Black Box" algorithms**

TIMELINE & MILESTONES



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